19. Scenario: You are a medical researcher investigating the effectiveness of a new drug in reducing blood pressure. You conduct a clinical trial with a sample of 50 patients who were randomly assigned to receive either the new drug or a placebo. After measuring their blood pressure levels at the end of the trial, you obtain the data for both groups. Now, you want to determine the confidence intervals for the mean reduction in blood pressure for both the drug and placebo groups. Question:"What is the 95% confidence interval for the mean reduction in blood pressure for patients who received the new drug? Also, what is the 95% confidence interval for the mean reduction in blood pressure for patients who received the placebo?

```python
import numpy as np
from scipy import stats
drug = np.array([12, 10, 15, 8, 11, 14, 9, 13, 10, 12])
placebo = np.array([5, 4, 7, 3, 6, 5, 2, 8, 4, 6])
drug_mean = np.mean(drug)
placebo_mean = np.mean(placebo)
drug_ci = stats.t.interval(
    0.95, df=len(drug)-1,
    loc=drug_mean,
    scale=stats.sem(drug)
)
placebo_ci = stats.t.interval(
    0.95, df=len(placebo)-1,
    loc=placebo_mean,
    scale=stats.sem(placebo)
)
print("Drug Group:")
print("Mean Reduction:", round(drug_mean, 2))
print("95% Confidence Interval:", drug_ci)
print("\nPlacebo Group:")
print("Mean Reduction:", round(placebo_mean, 2))
print("95% Confidence Interval:", placebo_ci)
```

```
Drug Group:
Mean Reduction: 11.4
95% Confidence Interval: (np.float64(9.8111130265089), np.float64(12.9888869734911))

Placebo Group:
Mean Reduction: 5.0
95% Confidence Interval: (np.float64(3.693942953076946), np.float64(6.306057046923054))
```

20. Scenario: You are a data scientist working for an e-commerce company. The marketing team has conducted an A/B test to evaluate the effectiveness of two different website designs (A and B) in terms of conversion rate. They randomly divided the website visitors into two groups, with one group experiencing design A and the other experiencing design B. After a week of data collection, you now have the conversion rate data for both groups. You want to determine whether there is a statistically significant difference in the mean conversion rates between the two website designs. Question: "Based on the data collected from the A/B test, is there a statistically significant difference in the mean conversion rates between website design A and website design B?"

```
import numpy as np
from scipy import stats
A = np.array([2.5, 3.0, 2.8, 3.2, 2.9, 3.1, 2.7, 3.0])
B = np.array([3.2, 3.5, 3.8, 3.4, 3.6, 3.7, 3.3, 3.5])
mean_A = np.mean(A)
mean_B = np.mean(B)
t_stat, p_value = stats.ttest_ind(A, B)
print("Mean Conversion Rate of Design A:", round(mean_A, 2))
print("Mean Conversion Rate of Design B:", round(mean_B, 2))
print("T-statistic:", round(t_stat, 4))
print("P-value:", round(p_value, 4))
if p_value < 0.05:
    print("There is a statistically significant difference.")
else:
    print("There is no statistically significant difference.")
```

```
Mean Conversion Rate of Design A: 2.9
Mean Conversion Rate of Design B: 3.5
T-statistic: -5.6125
P-value: 0.0001
There is a statistically significant difference.
```

21.Scenario: you are a scientist conducting research on rare elements found in a specific region. Your goal is to estimate the average concentration of a rare element in the region using a random sample of measurements. You will use the NumPy library to perform point estimation and calculate confidence intervals for the population mean.The rare element concentration data is stored in a CSV file named "rare_elements.csv," where each row contains a single measurement of the concentration. Question: write a Python program that allows the user to input the sample size, confidence level, and desired level of precision.

```python
import numpy as np
from scipy import stats
import math
data = np.array([12.5, 13.2, 11.8, 14.1, 12.9,
13.5, 12.1, 14.3, 13.0, 12.7,
13.8, 11.9, 12.6, 13.4, 14.0])
n = int(input("Enter sample size: "))
confidence = float(input("Enter confidence level (e.g., 95): "))
precision = float(input("Enter desired precision: "))
if n < 2 or n > len(data) or not 0 < confidence < 100 or precision <= 0:
    print("Invalid input.")
else:
    sample = np.random.choice(data, n, replace=False)
    mean = np.mean(sample)
    std = np.std(sample, ddof=1)
    t_value = stats.t.ppf((1 + confidence / 100) / 2, n - 1)
    margin = t_value * std / math.sqrt(n)
    print("Point Estimate:", round(mean, 4))
    print("Confidence Interval:",
          (round(mean - margin, 4), round(mean + margin, 4)))
    print("Margin of Error:", round(margin, 4))
    if margin <= precision:
        print("Desired precision achieved.")
    else:
        print("Desired precision not achieved. Increase sample size.")
```

```
Enter sample size: 10
Enter confidence level (e.g., 95): 90
Enter desired precision: 1.0
Point Estimate: 13.13
Confidence Interval: (np.float64(12.6809), np.float64(13.5791))
Margin of Error: 0.4491
Desired precision achieved.
```

22.Scenario: Imagine you are an analyst for a popular online shopping website. Your task is to analyze customer reviews and provide insights on the average rating and customer satisfaction level for a specific product category. Question: You will use the pandas library to calculate confidence intervals to estimate the true population mean rating. You have been provided with a CSV file named "customer_reviews.csv," which contains customer ratings for products in the chosen category.

```
!pip -q install pypdf
import numpy as np
from scipy import stats
from pypdf import PdfReader
from google.colab import files
import re
uploaded = files.upload()
filename = next(iter(uploaded))
reader = PdfReader(filename)
text = ""
for page in reader.pages:
    text += page.extract_text() or ""
ratings = re.findall(r'\b[1-5](?:\.\d+)?\b', text)
ratings = np.array([float(x) for x in ratings])
if len(ratings) < 2:
    print("Not enough ratings found in the PDF.")
else:
    mean = np.mean(ratings)
    std = np.std(ratings, ddof=1)
    n = len(ratings)
    t_value = stats.t.ppf(0.975, n - 1)
    margin = t_value * std / np.sqrt(n)
    print("Number of Ratings:", n)
    print("Average Rating:", round(mean, 2))
    print("95% Confidence Interval:",
          (round(mean - margin, 2), round(mean + margin, 2)))
```

```
if mean >= 4:
    print("Customer Satisfaction: High")
elif mean >= 3:
    print("Customer Satisfaction: Moderate")
else:
    print("Customer Satisfaction: Low")
```

```
405.1/405.1 kB 7.2 MB/s eta 0:00:00
Choose Files Customer P…Ratings.pdf
Customer Product Ratings.pdf(application/pdf) - 47713 bytes, last modified: 10/20/2026 - 100% done
Saving Customer Product Ratings.pdf to Customer Product Ratings.pdf
Number of Ratings: 26
Average Rating: 3.96
95% Confidence Interval: (np.float64(3.54), np.float64(4.38))
Customer Satisfaction: Moderate
```

23.Scenario: You are a researcher working in a medical lab, investigating the effectiveness of a new treatment for a specific disease. You have collected data from a clinical trial with two groups: a control group receiving a placebo, and a treatment group receiving the new drug.Your goal is to analyze the data using hypothesis testing and calculate the p-value to determine if the new treatment has a statistically significant effect compared to the placebo. You will use the matplotlib library to visualize the data and the p-value.

```
import numpy as np
from scipy import stats
control = np.array([20, 22, 19, 24, 21, 23, 20, 22, 21, 19])
treatment = np.array([15, 17, 14, 18, 16, 15, 17, 16, 14, 18])
t_stat, p_value = stats.ttest_ind(control, treatment)
print("Control Group Mean:", np.mean(control))
print("Treatment Group Mean:", np.mean(treatment))
print("T-statistic:", round(t_stat, 4))
print("P-value:", round(p_value, 4))
if p_value < 0.05:
    print("Reject the null hypothesis.")
    print("The treatment has a statistically significant effect.")
else:
    print("Fail to reject the null hypothesis.")
    print("The treatment has no statistically significant effect.")
```

```
Control Group Mean: 21.1
Treatment Group Mean: 16.0
T-statistic: 7.2205
P-value: 0.0
Reject the null hypothesis.
The treatment has a statistically significant effect.
```

24.Question: K-Nearest Neighbors (KNN) Classifier You are working on a classification problem to predict whether a patient has a certain medical condition or not based on their symptoms. You have collected a dataset of patients with labeled data (0 for no condition, 1 for the condition) and various symptom features. Write a Python program that allows the user to input the features of a new patient and the value of k (number of neighbors). The program should use the KNN classifier from the scikit-learn library to predict whether the patient has the medical condition or not based on the input features.

```python
from sklearn.neighbors import KNeighborsClassifier
X = [[1, 0, 1],[0, 1, 0],[1, 1, 1],[0, 0, 0],
    [1, 0, 0],[0, 1, 1]]
y = [1, 0, 1, 0, 1, 0]
k = int(input("Enter the value of k: "))
if k < 1 or k > len(X):
    print("Invalid value of k")
else:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X, y)
    fever = int(input("Enter fever (0 or 1): "))
    cough = int(i
    fatigue = int
    patient = [[f
    prediction =
    if prediction[0] == 1:
        print("Patient has the medical condition.")
    else:
        print("Patient does not have the medical condition.")
```

KNeighborsClassifier: model

sklearn.neighbors._classification.KNeighborsClassifier instance

```
Enter the value of k: 3
Enter fever (0 or 1): 1
Enter cough (0 or 1): 1
Enter fatigue (0 or 1): 1
Patient has the medical condition.
```